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夸克英语笔记

Lesson 14 The Butterfly Effect

蝴蝶效应 祈使句 并列句

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forecast / ˈfɔːkɑːst / n. 预报
blizzard / ˈblɪzəd / n. 暴风雪
multiply * / ˈmʌltɪplaɪ / v. 增加
turbulent / ˈtɜːbjələnt / adj. 狂暴的
squall / skwɔːl / n. 暴风
grid / grɪd / n. 坐标方格
humidity / hjuːˈmɪdəti / n. 湿度
Princeton / ˈprɪnstən / n. 普林斯顿（美国城市）
fluctuation / ˌflʌktʃuˈeɪʃ(ə)n / n. 起伏，波动
speculative / ˈspekjələtɪv / adj. 推测的
deteriorate / dɪˈtɪəriəreɪt / v. 变坏，恶化
cascade / kæˈskeɪd / v. 瀑布似的落下
dust devil / ˈdʌst devl / n. 小尘暴，尘旋风
eddy / ˈedi / n. 旋涡
sensor / ˈsensə(r) / n. 传感器
meteorologist / ˌmiːtiəˈrɒlədʒɪst / n. 气象学家
New Jersey / njuː ˈdʒɜːzi / n. 新泽西（美国州名）
deviation / ˌdiːviˈeɪʃn / n. 偏差

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※ The Butterfly Effect

『**butterfly** 蝴蝶；蝶泳；举止轻浮的人；追求享乐的人』

- She is known as a **social butterfly.** 社交蝴蝶（指举止轻浮或追求享乐的人）、交际花
- The children chased a beautiful yellow **butterfly** in the garden.
- He is an expert in the **butterfly** stroke.

have butterflies (in one's stomach) 忐忑不安	• It was the morning of the World Cup Final and most of the players had **butterflies (in their stomach)**. • I always **have butterflies** before a major presentation. • She confessed to having **butterflies in her stomach** before her wedding.

『 **chaos theory** 混沌理论 』

> **Chaos theory** suggests that small changes in initial conditions can lead to large-scale and unpredictable variation in the future state of the system.

- The **Butterfly Effect** is a frequent metaphor for one aspect of **chaos theory**.
- The principle illustrates how minor initial differences can be amplified, leading to widely diverging outcomes.

※ Beyond two or three days, the world's best weather forecasts are speculative, and beyond six or seven they are worthless.

『 **Beyond** prep. after ...; later than ... 』

- **Beyond two or three days**, the world's best weather forecasts are speculative.

after... 在...之后	**After three days**, the predictions become uncertain.
later than... 晚于...	Any forecast **later than two days** is unreliable.
Exceeding v.超过（数量）；超越，越出（权限）	**Exceeding two or three days**, the accuracy drops significantly.

『 **speculative** 推测性的；投机的 』

- If so, they would gain **speculative** profits. 这样的话，它们就有可能获得投机性收益。
- Liquidity inflow fuelled **speculative** demand. 资金流入刺激投机性需求。
- Venture capital transmogrified into **speculative** fever. 风险资本改头换面进入投机热。

be based on speculation / conjecture 基于推测	The reports are **based on speculation**, not facts.
be based on guesswork / guess work 基于猜测	Many long-range forecasts are simply **based on guesswork.** 只是基于猜测

『 **worthless** 没有价值的(valueless、useless、...of no earthly use、totally useless) 』

- They finally found a small gold coin which was almost **worthless**.
- Critics say his paintings are worthless. 批评家说他的画毫无价值。
- Constant rejections made him feel worthless. 不断地遭到拒绝使他觉得一无是处。

valueless / useless 没有价值的/无用的	The old currency is now **valueless**. / The advice proved to be **useless**.

of no earthly use 毫无用处（语气重）	That broken machine is **of no earthly use** to anyone.
totally useless 完全无用的	The new feature was **totally useless** for its intended purpose.

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※ The Butterfly Effect is the reason.

> ~~The reason is because of the Butterfly Effect.~~ X reason和because重复

『reason for / behind 背后的原因』

- The **Butterfly Effect is the reason for / behind it**. 替换
- What is the **reason for** his sudden departure?
- She explained the **reason behind** her decision.

accounts for 是...的原因；解释	The Butterfly Effect **accounts for** it.蝴蝶效应解释了它
underly 是...的基础/原因	The Butterfly Effect **underlies** it.蝴蝶效应是其背后的原因
The reason is... 原因是...	**The reason is** the Butterfly Effect.原因是蝴蝶效应
The reason lies in... 原因在于...	**The reason lies in** the Butterfly Effect.原因**在于**蝴蝶效应

※ For small pieces of weather -- and to a global forecaster, small can mean thunderstorms and blizzards -- any prediction deteriorates rapidly.

『mean v.be likely to result in ... 导致、意味着...』

- The merger will **mean** the eventual closure of the company's Shanghai Office.将意味着公司上海办事处的最终关闭
- This delay will **mean** a lot of extra work for everyone. 将意味着大量的额外工作
- It would almost certainly **mean** the end of NATO. 这的确几乎导致北大西洋公约组织的终结。

『deteriorate vi. 恶化，变坏；vt. 恶化』

- ... any prediction **gets worse / degenerates / becomes inaccurate** rapidly.
- The situation is expected to **deteriorate** in the coming years. 预计这种局面在未来数年还会恶化。
- Most importantly, buildings need love if they are not to **deteriorate**. 最重要的是，建筑需要爱护才不会老化。

gets worse / degenerates 恶化，变坏	• His condition **gets worse** every day. 每况日下 • The quality of the film **degenerates** in the second half.
becomes inaccurate 变得不准确	Beyond this point, the data **becomes inaccurate** rapidly.迅速变得不准确

※ Errors and uncertainties multiply, cascading upward through a chain of turbulent features, from dust devils and squalls up to continent-size eddies that only satellites can see.

『 **multiply** v.to increase by a large amount or number, or to make sth. do this （使）大幅增加 』

- Within the last 10 years, both sales and profits have **multiplied.**大幅增加
- Smoking **multiplies** the risk of heart attacks.成倍增加了心脏病发作的风险
- This will **multiply** our chances of success. 这将增加我们成功的机会。

snowball （像滚雪球般）迅速增加	Unemployment **snowballed** at the beginning of 1980s.
proliferate 激增，扩散	Small businesses have **proliferated** in the last ten years.
mushroom 迅速增长，如雨后春笋般涌现	Trade between the two countries has **mushroomed**.
explode 爆炸式增长	China's population **exploded** after 1949.

『 **a chain of...** 一连串的 』

- Life is **a chain of** moments of enjoyment, not only about survival. 生活是一串串的快乐时光，我们不仅仅是为了生存而生存。
- She had a **train of** thought.一连串的想法
- He suffered **a string of** defeats.一连串的失败
- We endured **a succession of cold days.**一连串的寒冷日子

a series of ... 一连串的；一系列的	The scientist conducted **a series of** experiments to test the hypothesis. 科学家进行了一系列的实验来检验这个假设。
a train of ... 一连串的；一系列的(尤指事件、想法)	The news started **a train of** thoughts about her childhood. 这条新闻引发了一连串关于她童年的想法。

a string of ... 一连串的；一系列的 (尤指相似或相关的事件)	The store had **a string of** bad luck, including a fire and two robberies. 这家商店遭遇了一连串的厄运，包括一场火灾和两次抢劫。
a succession of ... 连续的；接连不断的	The team won the championship after **a succession of** victories. 在一连串的胜利之后，该团队赢得了冠军。
a progression of ... 一系列的；进展过程中的	The painting shows **a progression of** colors from light blue to dark violet. 这幅画展示了从浅蓝色到深紫色的一系列色彩变化。

『 **feature** n. characteristic 特征（指人为五官、指物为特点） 』

- The house has many appealing **features.** 吸引人的特征
- The climate is a key **feature** of this region.

『 **from A up to B** 从A一直到B 』

- The survey covers everyone **from teenagers up to the elderly.** 从青少年一直到老年人
- The prices range **from $10 up to $100.** 从10美元一直到100美元

from A to B 从A到B	How to improve your travel **from** home **to** work? 如何改进你从家到办公地点的行程?

工作原理 (笔记最后修改于:2025-11-11 11:33:53)

※ The modern weather models work with a grid of points of the order of 60 miles apart, and even so, some starting data has to be guessed, since ground stations and satellites cannot see everywhere.

『 **work** v.(of a machine, device, etc) function; operate （指机器、设备等）运作，运转 』

- This machine **works by electricity.** 靠电力运作
- The delete key doesn't **work.** 不起作用

『 **grid** n.a pattern of straight lines that cross each other to form squares （地图上的）坐标方格 』

- Streets laid out on a **grid** pattern.
- The city has **streets laid out on a grid pattern.** 街道按网格模式布置
- We located the position on the **grid.** 坐标方格

『 **of / in / on the order of...** 大约、大概(approximately) 』

- a figure **in the order of** $7 million
- Her salary is **of the order of** $150 a week.
- And the available volume is **on the order of** meters cubed. 整个可用的空间体积有立方米的数量级。
- They borrowed something **in the order of** $10 million. 他们大约借了一千万美元。
- But is this simply part **of the order of** Nature? 但这是自然法则吗?

- The house cost him something **on the order of** $200,000.

in the neighborhood / vicinity of... 大约、大概	The house cost him something **in the neighborhood / vicinity of** $200,000.
approximately 大约、大概	The house cost him **approximately** $200,000.大约20万美元

『 **starting** adj. beginning 最初的 』

- ...some **starting data**一些初始数据 has to be guessed...

initial 最初的；字首的	at the **initial** stage / phase.
incipient 初期的；初始的	• an **incipient** tumor. • the **incipient** light of day
opening 开始的；开篇的	• his **opening** remarks. • the **opening** scene of the movie.电影的开篇场景
preliminary 预备性的；初步的	a **preliminary** exam / contest.

※ But suppose the earth could be covered with sensors spaced one foot apart, rising at one-foot intervals all the way to the top of the atmosphere.

『 **space** v.将...均匀隔开 』

- **Space** your workouts two days apart to avoid sore muscles.
- Bean plants should **be spaced** three inches apart.应该间隔三英寸
- The columns are **spaced evenly** along the wall.

interval	（两事件中的）间隔时间	..., and by measuring the time **interval** between the taps and the receipt of the echoes the depth of the sea at that point can be calculated.
	（物体之间的）间隔空间	They planted trees in the **intervals** between the houses.

『 **all the way** 一直... 』

- He ran **all the way** home.他一路跑回家
- Eventually the turtle has been butted **all the way** down to the floor of the tank.
- The noise could be heard **all the way** down the street.在整条街上都能听到

※ Suppose every sensor gives perfectly accurate readings of temperature, pressure, humidity, and any other quantity a meteorologist would want.

『reading 读数』

- read the meter 查表
- The nurse took the patient's vital **readings.** 生命体征读数
- We need to **take the meter reading** before leaving.

※ (Suppose) Precisely at noon an infinitely powerful computer takes all the data and calculates what will happen at each point at 12.01, then 12.02, then 12.03...

『Precisely at noon 正午时分(12点整)』

- **Precisely at two-thirty** someone knocked on the door. Lamar opened the door. 两点半的时候，有人敲门。拉马尔开了门。
- It is **precisely at this juncture** that the boss finds himself in double jeopardy. 恰恰在这个关键时刻，上司发现自己处于进退两难的境地。
- It is difficult to measure the density of rocks **precisely at every part** of the mountain region. 而要准确的测定各部分岩石密度又是十分困难的。

at high noon 正午时分	The duel was set for **at high noon.** 正午时分
at 12 o'clock sharp 12点整	The meeting starts **at 12 o'clock sharp.** 12点整
at punctually 12 o'clock 准时12点	Please arrive **at punctually 12 o'clock.** 准时12点

『infinite 无限的，无穷的；无数的；极大的』

- The universe is **infinite.** 无限的
- The technology has been working well for me at the office, but there are **infinite** applications. 这项技术在我的办公室里运行得很好，但它有无限的应用。
- Never lose your heart and continue to explore life since it is full of **infinite** possibilities. 永远不要灰心，要继续探索生活，因为它充满了无限的可能。

finite 有限的；限定的（反义词）	Human knowledge is **finite.** 有限的
infinitive adj. 原形的，不定式的	It is wrong to ever split an **infinitive.** 将不定式分割永远是错误的。
definite adj. 一定的；确切的	Our ideas began to crystallize into a **definite** plan. 我们的想法开始形成了一个明确的计划。

indefinite 不确定的；无限的；模糊的（形近词）	• The project was postponed for an **indefinite period.** 不确定的时期 • an **indefinite** answer
definitive 最终的；确定的（形近词）	a **definitive verdict.** 最终裁决

『 **calculate** 计算 』

- Can you **calculate** the total cost of these items? 你能计算出这些物品的总成本吗?
- Can you **work out** the total cost?

work out 计算出 (phr. v.)	We need to **work out** the exact amount before we proceed. 在开始之前，我们需要计算出准确的金额。
figure out 计算出；想出 (phr. v.)	I can't **figure out** how much we owe them. 我算不出来我们欠他们多少钱。
compute 计算 (v.)	The system can **compute** complex equations in seconds. 该系统可以在几秒钟内计算出复杂的方程式。
reckon 计算；估计 (v.)	The inflation is **reckoned** to be 10%. 通货膨胀被估算为10%。

现实情况 (笔记最后修改于:2025-11-11 11:53:48) 4

※ The computer will still be unable to predict whether Princeton, New Jersey, will have sun or rain on a day one month away.

『 **unable to do sth.** 不能做某事 』

- The team was **unable to score** in the second half.
- I was **unable to attend** the conference due to illness.

『 **one month / week / year away** 一个月以后 』

- Ross and Emily set a wedding date just **one month away**. 罗斯和艾米丽计划下个月结婚。
- The election is only **one week away**. There are rallies all over town tonight. 只剩一个星期就要选举了。今晚全城各地都有群众大会。
- London's games are a mere **two years away**. 距伦敦奥运会开幕仅剩两年的时间。
- Cant believe we are **two days away** from 2027. 不敢相信离2027还有两天了.

※ At noon the spaces between the sensors will hide fluctuations that the computer will not know about, tiny deviations from the average.

『 **fluctuations** 波动，起伏 』

- We observed constant **fluctuations in the stock market**.
- Temperature **fluctuations** can affect the experiment's results.

『**deviations from the average** 与平均值的偏差』

- Any **deviation from the standard procedure** must be reported.
- The quality control checked for **deviations from the norm**.

※ By 12.01, those fluctuations will already have created small errors one foot away.

『**already have created** 已经造成了（将来完成时）』

- By next month, I will **already have finished** the entire project.
- By the time he arrives, they will **already have left**.

※ Soon the errors will have multiplied to the ten-foot scale, and so on up to the size of the globe.

『**multiplied** 大幅增加』

- Our problems have **multiplied** since last year. 自去年以来，我们的问题成倍增加。
- It's an administrative sham of the kind that has **multiplied** over the last 30 years. 这是一种行政骗局，在过去30年里该骗局不断激增。

snowballed to... （像滚雪球般）迅速增加到...	The rumours will have **snowballed to the ten-foot scale.** 谣言将迅速增加到十英尺的规模
proliferated to... 激增到...	The problems will have **proliferated to the ten-foot scale.** 问题将激增到十英尺的规模
rocketed to... 猛增到...	The numbers will have **rocketed to the ten-foot scale.** 数字将猛增到十英尺的规模
soared to... 飙升到...	The costs will have **soared to the ten-foot scale.** 成本将飙升到十英尺的规模

『**scale** 程度、规模、范围』

- The **scale** of the fighting is almost unimaginable. 战斗的规模几乎难以想像。
- There is no precedent for a disaster of this **scale**. 这种规模的灾难是空前的。
- What began as a minor scuffle turned into **a full-scale** riot. 开始的小冲突演变成了大规模的暴乱。

on a small / limited scale 小规模/有限规模	The experiment was conducted **on a small scale.** 以小规模进行

on a large / grand scale 大规模	The changes were implemented **on a grand scale.** 以大规模实施
on a / an colossal / enormous / gigantic scale 特大规模	They planned a building project **on an enormous scale.** 以特大规模进行
on a national / nationwide scale 全国规模	The campaign launched **on a national scale.** 在全国规模启动
on a / an global / international / world / worldwide scale 全世界规模	The crisis affected markets **on a global scale.** 影响了全球规模的市场

『and so on 等等』

- We discussed everything—when to go, what to see **and so on**. 我们什么都商量过了—什么时候走、看什么等等。
- Don't forget about texts, emails, **and so on**. 不要忘记短信、邮件等等。
- There is tutoring, sports clubs **and so on**. 有补习班、体育俱乐部等等。

and so forth 等等	We discussed music, movies, and **so forth.** 等等
etc. (et cetera) 等等	The list includes apples, oranges, pears, **etc.** 等等
and the like 以及类似事物	He collects stamps, coins, **and the like.** 以及类似事物
and what not 以及诸如此类的事物	They sell crafts, jewelry, **and what not.** 以及诸如此类的事物

『global size 全世界规模』

- **Global** regions are weighted by the **size** of their GDP. 而衡量世界国家地区规模的标准是GDP。
- That is the **size of the global** telecom market. 而这就是全球电信市场的规模。
- Still, while the **size** and reach of **global** illicit industries can seem overwhelming, there are signs of hope. 尽管全球非法产业的规模越来越大，分布越来越广，我们依旧能够看到希望。

global scale 全世界规模	This environmental problem requires action on a **global scale.** 这个环境问题需要全世界规模的行动。
international scale 国际规模	The conference brought together experts working on an **international scale.** 这次会议汇集了在国际规模上工作的专家。
world scale 世界规模	The company is now operating production facilities on a **world scale.** 该公司现在正在以世界规模运营生产设施。

worldwide scale 全球规模	The new technology is expected to be adopted on a **worldwide scale**. 这项新技术预计将在全球规模上被采纳。



Play it by ear.

随机应变

Idioms